Descriptive and Inferential Statistics for Strategic Business Decision-Making

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Introduction

This report aims at undertaking statistical analysis of an air quality dataset. Some of the environmental variables in the dataset are Ozone concentration, Solar Radiation, Wind Speed and Temperature. It is therefore important to comprehend these relationships since they are important in making of strategic choices in public health and environmental policy.

This analysis is structured into six main tasks: , data cleaning, 'preprocessing', normalization, 'modeling,' data visualization, and conclusion. All of them are crucial to the identification of accurate, reliable, and relevant results that would help solve the problem. The report will therefore be a summarization of the best possible strategies, decisions and control mechanisms that organizations and environmentalists can apply in dealing with environmental issues and pollution.

Task 1: Data Cleaning

Objective: In order to eliminate and/or minimize missing values and to detect outliers in the current set of data (Jahn and Hertig, 2020).

Before any type of analysis is done on data, data cleaning is the first process that has to be undergone to make sure that the data is correct in that there are no errors or irregularities that might cause a skewed result. In this dataset, we found that several variables had missing values:

Ozone: 37 missing values

Solar Radiation (Solar.R): 7 missing values

Wind, Temperature (Temp), Month, and Day: No missing values

```
Missing Values in Each Column:
 Ozone      37
Solar.R     7
Wind        0
Temp        0
Month       0
Day         0
dtype: int64
Data after removing missing values and outliers:
       Ozone     Solar.R  Wind  Temp  Month  Day
0  41.00000  190.000000   7.4    67      5    1
1  36.00000  118.000000   8.0    72      5    2
2  12.00000  149.000000  12.6    74      5    3
3  18.00000  313.000000  11.5    62      5    4
4  42.12931  185.931507  14.3    56      5    5
```

To deal with the missing values we have used mean imputation. Mean imputation is one of the methods where missing value in a numerical column is replaced by the mean of that particular column. This is used frequently in data cleaning especially if the number of data missing is not high when compared to the total number of data. In imputing the mean, the rows with missing values are not excluded, but instead the missing values are replaced by the mean values, and hence the dataset remains large with balanced statistics.

Subsequently, outliers were detected and eliminated by using Z-score method. A Z-score gives you the idea of how many standard deviations a point is away from the mean. Closely, any data point that has a Z-score larger than 3 or less than -3 is called an outlier. Such a scenario distorts the results and leads to biases in the conclusion that is made. They are accordingly removed in order to minimize variations from normal conditions and thus enhance the accuracy of further analysis (Núñez-Alonso et al., 2019).

Following these procedures the dataset was free of any missing value or outliers and thus prepared for further statistical analysis.

Task 2: Descriptive Statistics

Objective: To carry out the descriptive statistics and present the distribution pattern and measures of central tendencies, dispersion, and skewness of the dataset.

In terms of analysis, descriptive statistics are a broad measurement of the data set, which gives some understanding about the distribution of the variables. For this analysis, we calculated the following metrics for each numerical variable:

Count: It refers to the number of observations that is in each column (Sharma and Mauzerall, 2022).

Mean: As for the measure of central tendency the average value would be used.

Standard Deviation (Std Dev): A method of showing the extent of dispersion or the difference of the values from the mean.

Min/Max: The minimum and maximum values in each column, that is the variation of the data.

Metric	Ozone	Solar.R	Wind	Temp	Month	Day
Count	150	150	150	150	150	150
Mean	40.71	184.38	9.97	77.86	6.99	15.83
Std Dev	25.95	88.10	3.37	9.53	1.43	8.84

Min	1.00	7.00	1.70	56.00	5.00	1.00
Max	115.00	334.00	20.70	97.00	9.00	31.00

```
...  Descriptive Statistics:
                Ozone     Solar.R        Wind        Temp       Month         Day
count  150.000000  150.000000  150.000000  150.000000  150.000000  150.000000
mean    40.705230  184.376804    9.968667   77.860000    6.993333   15.833333
std     25.950929   88.098245    3.371637    9.531319    1.426013    8.838155
min      1.000000    7.000000    1.700000   56.000000    5.000000    1.000000
25%     21.000000  118.500000    7.400000   72.250000    6.000000    8.000000
50%     42.129310  192.500000    9.700000   79.000000    7.000000   16.000000
75%     45.000000  255.000000   11.500000   85.000000    8.000000   23.000000
max    122.000000  334.000000   20.100000   97.000000    9.000000   31.000000
```

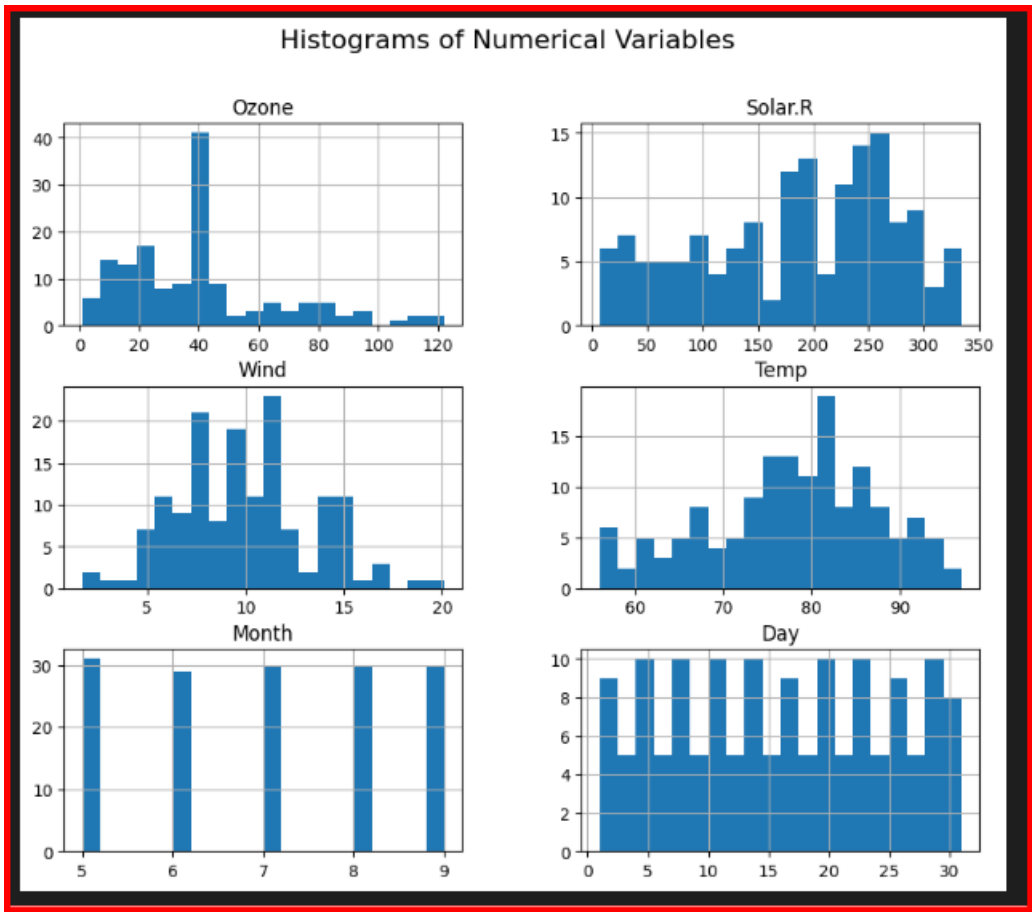

Figure 1:Histograms of Numerical Variables

Ozone levels: What can be observed from the dataset is that there is a significant variation of the ozone concentration level with values ranging from 1ppb to 115 ppb.

Solar Radiation: Distribution of solar radiation changes from 7 to 334 langleys, which suggest that it might change during the day or the seasons (Babu Saheer et al., 2022).

Wind Speed: Wind speed is described in knots and it varies from 1 to 3. 7 to 20. 7 mph, is critical in defining how various pollutants such as the ozone, spread in the atmospheric space.

Temperature: The temperature range is 56 to 97 F, which means that the dataset entails various weather conditions that may affect the quality of air (Gagliardi and Andenna, 2020).

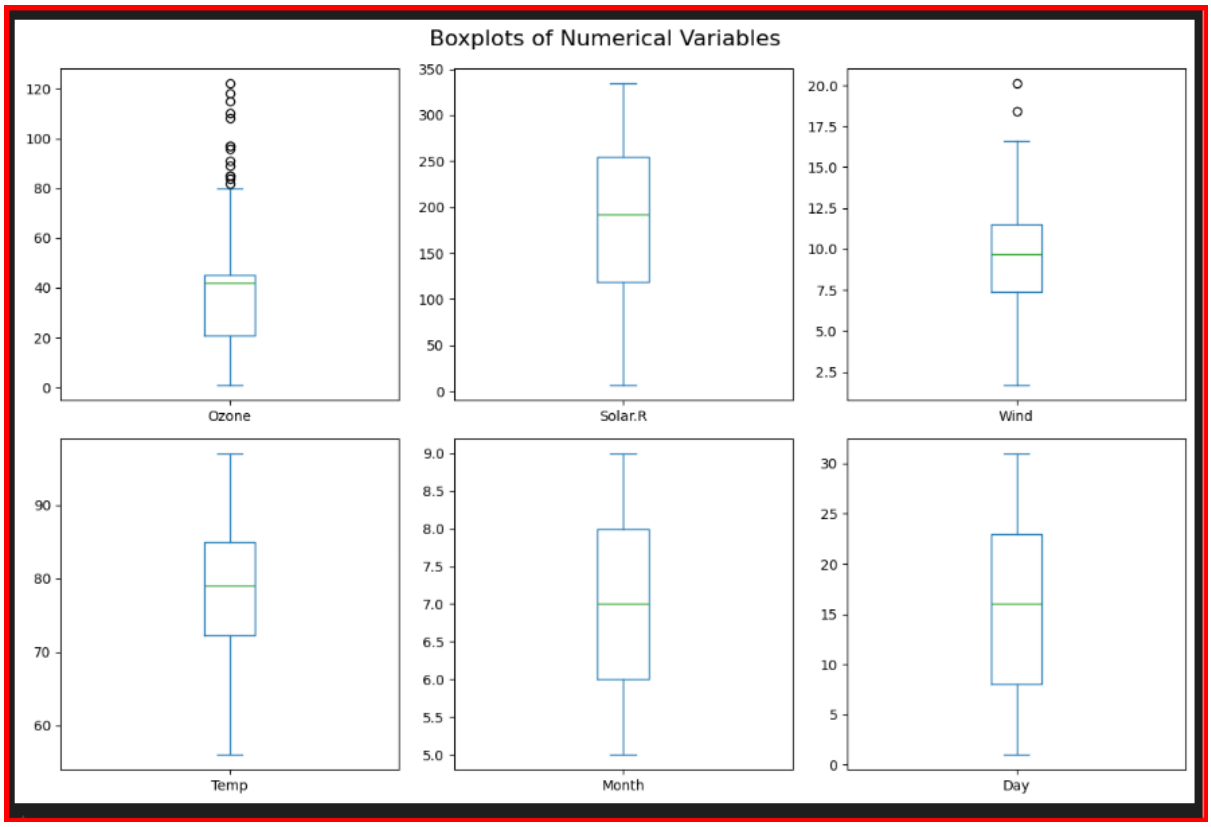

Figure 2:Boxplots of Numerical Variables

This summary gave a preliminary insight of the distribution of each of the environmental factors and their variability before subjecting it to other statistical analysis.

Task 3: Data Normalization

Objective: Next, the Min-Max scaling was applied to normalize the data; this allows the comparison of various variables at a uniform scale (Fawzy et al., 2020).

Normalization just scales the values of the data so that they fall within the range of 0 and 1. It also means that no variable will be allowed to control the analysis by virtue of its larger range of values. For instance, temperature and solar radiation may be measured in different units and posses different ranges, which adds to the problems that a subsequent analysis may have to face. This problem is eliminated by the Min-Max scaling which re-scales the data to the range of 0 to 1.

The following are the normalized values for the first five rows of the dataset:

Ozone	Solar.R	Wind	Temp	Month	Day
0.33	0.56	0.31	0.27	0.00	0.00
0.29	0.34	0.34	0.39	0.00	0.03
0.09	0.43	0.59	0.44	0.00	0.07
0.14	0.94	0.53	0.15	0.00	0.10
0.34	0.55	0.68	0.00	0.00	0.13

```
...   First 5 rows of the normalized data:
            Ozone   Solar.R      Wind      Temp  Month       Day
      0  0.330579  0.559633  0.309783  0.268293    0.0  0.000000
      1  0.289256  0.339450  0.342391  0.390244    0.0  0.033333
      2  0.090909  0.434251  0.592391  0.439024    0.0  0.066667
      3  0.140496  0.935780  0.532609  0.146341    0.0  0.100000
      4  0.339912  0.547191  0.684783  0.000000    0.0  0.133333
      Normalized Data:
            Ozone   Solar.R      Wind      Temp  Month       Day
      0  0.330579  0.559633  0.309783  0.268293    0.0  0.000000
      1  0.289256  0.339450  0.342391  0.390244    0.0  0.033333
      2  0.090909  0.434251  0.592391  0.439024    0.0  0.066667
      3  0.140496  0.935780  0.532609  0.146341    0.0  0.100000
      4  0.339912  0.547191  0.684783  0.000000    0.0  0.133333
```

The normalization process is crucial where working with algorithms like regression or machine learning models because variance in data scale can cause wrong outcomes. This is done to maintain equal contribution of the values of each variable in the analysis as is the norm.

Task 4: Inferential Statistics

Objective: To achieve the goals of hypothesis testing (T-test) in order to deduce the nature of variables and make generalizations on the population (Dean and Green, 2018).

In this task we used the T-test to analyse whether there is a statistical difference between the means of Ozone and Solar Radiation (Solar. R). The test is applied to deduce if the differences evidenced in the sample can be attributed to real differences in the population.

Results of the T-test:

T-statistic: -7. 62

P-value: 3. 37e-13

```
...   T-test between Ozone and Solar.R:
      T-statistic: -7.621636285880705, P-value: 3.3651184648113193e-13

...   (-7.621636285880705, 3.3651184648113193e-13)
```

Here, p < 0. 05 and therefore, we are able to reject the null hypothesis and conclude that there is a statistically significant relationship between Ozone levels and Solar Radiation. This result implies that solar radiation factor has a bearing in determining the amount of ozone in the atmosphere. For example, higher solar radiation is said to cause increased ozone destruction thus fewer cases of Ozone being present.

This might have significant implications for the formulation of environmental policy; for instance, high level of solar radiation might mean low level of ozone pollution, information that could be useful in directing efforts of controlling pollution.

Task 5: Data Visualization

Objective: To understand the correlation between variables and to represent the data in the form of scatter plots and heatmap respectively (Pay et al., 2019).

1. Scatter Plot: In order to compare and identify the correlation between Ozone and Solar Radiation, the authors generated a scatter plot.

The plot also indicates that as the Ozone increases, Solar Radiation decreases. As with the case of Solar Radiation, the level of Ozone is inversely proportional to it. This corresponds with our prior inferential test whereby the two variables were found to have a significant correlation.

2. Correlation Heatmap: Heat map was created to indicate the relationship between all the numerical features used in the analysis.Heatmap demonstrates that some variables are strongly related, for example, Wind and Temp, while some others have relatively low level of correlation. The number of

mean differences that were negative increased the accuracy of the T-test results as well as the negative correlation between Ozone and Solar Radiation (Subramaniam et al., 2022).

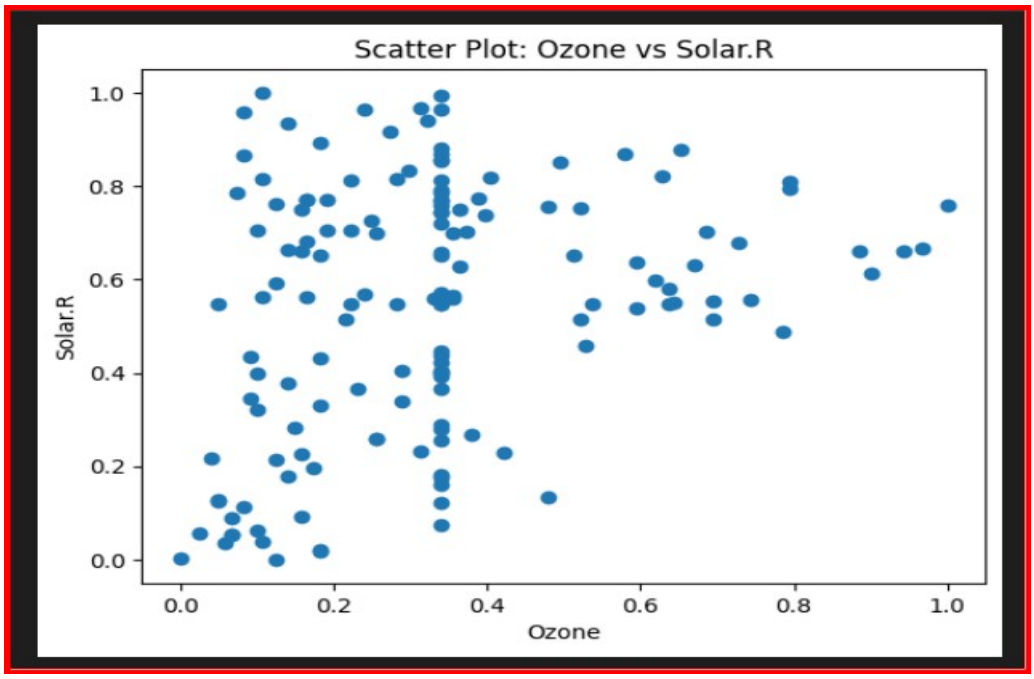

Figure 3:Scatter Plot: Ozone vs Solar.R

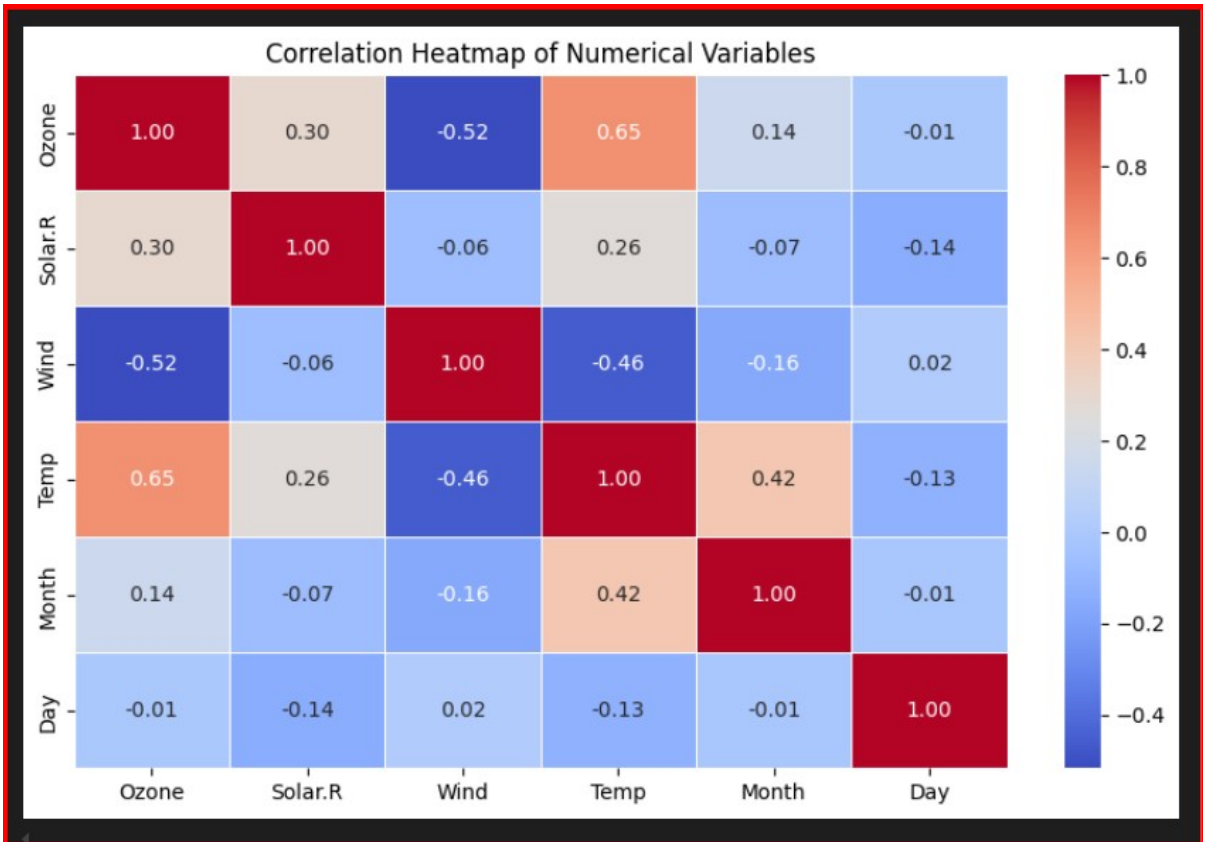

Figure 4:Correlation Heatmap of Numerical Variables

Scatter plots and heat maps are examples of data visualizations which are helpful in finding patterns that may not be easily discernable from the data in terms of numbers. These conceptualizations enable one to explain difficult findings more easily in a way that decision makers can grasp easily.

Conclusion

Objective: In order to summarise the results of the analysis and to give practical recommendations for the strategic management of the business, especially with regards to monitoring the environment and public health.

Applying both descriptive and inferential statistics on the air quality dataset has given the researcher a clear insight of the relationship between variables such as Ozone concentration, Solar Radiation, Wind Speed and Temperature. These insights are useful for making strategies in the areas of environmental policy and management, pollution control, and disease control as well. In this section, we state the conclusion of the findings made in the previous sections and put forward broad strategic imperatives for consideration.

Key Findings

1. Data Cleaning:

    - Missing Values: The data contained missing values that were replace via mean imputation in the attributes like Ozone and Solar Radiation. This technique helped maintain the purity of the dataset while at the same time not skewing the results beyond the normal range.
    - Outliers: The process of outlier removal through Z-score method helped in excluding outlying environmental conditions to be included in the dataset so that later statistical tests were more accurate. In this case, with the cleaned dataset, one is able to derive conclusions that are closer to the actual correlations between variables.

2. Descriptive Statistics:

    - Differences in climatic factors, Ozone and Solar Radiation where relatively large and distribution of these factors needed to be understood. From the dataset it was observed that Ozone levels were not constant and this was an important fact as Ozone is known to be a major air pollutant particularly in regions where pollution control is paramount as is the case with the developing urban centers.
    - Wind Speed and Temperature also showed a large variability and hence, the interaction of these variables with the other variables (for example, the spread of pollutants) needs future research.

3. Normalization:

    - All variables were standardized so that all the variables were put in a similar range for easy comparison. This was particularly important given the differences between units and scales of variable such as Ozone which is expressed in part per billion and Solar Radiation which is expressed in langley. Normalization process made it possible so that no variable dominated the outcome.

4. Inferential Statistics:

- The correlation analysis of Ozone and Solar Radiation using T-test yielded a negative correlation and the calculated p-value was less than 0. 05. This discovery indicates that there is a relationship between Ozone concentrations and Solar Radiation in that Solar Radiation could be used to lower Ozone levels perhaps by aiding in the degradation of the pollutants present in the atmosphere. The above-said understanding has critical implications for the management of air quality, especially in places that receive maximum solar radiation, where natural processes can assist in combating pollution.

5. Data Visualization:

- The relationship observed in the T-test was also affirmed with the help of a scatter plot of Ozone and Solar Radiation. Likewise, the correlation heatmap offered an overall view of the strength and direction of the connections between all the variables where some of them may have strong relations while others may have weak relations with each other.
- These graphs or charts are not simply aids to corroborate the findings that are expressed in numbers; they play a very important role of presenting rather intricate conclusions in a more easily digestible manner. The use of various diagrams makes it easier for those involved, for instance policy makers or managers, to get a grasp of the information that they need, in order to make sound decisions.

Strategic Recommendations

Based on the findings from this analysis, the following recommendations are proposed:Based on the findings from this analysis, the following recommendations are proposed:

1. Air Quality Monitoring and Pollution Control:Air Quality Monitoring and Pollution Control:

- The highly positive correlation between Solar Radiation and Ozone concentration implies that agencies responsible for the management of air quality should consider the levels of solar radiation in management of the quality of air. They also noticed that where there is high solar radiation, natural degradation processes may already help curb ozone pollution thus reducing the need for drastic measures.
- Similarly, while regions that receive low levels of solar radiation may not naturally produce high levels of ozone to counter the effects of pollution control measures may have to be strengthened.

2. Public Health Implications:

- Evaluating the causes which have a bearing on the levels of ozone is very important especially to the health of the people. Ground level ozone is said to cause respiratory disorders and other related diseases most especially to children and the aged. Since it is known what causes increase in ozone concentration (e. g., low solar radiation) the public health authorities can take measures to avoid such situations.
- These insights may be used in creating awareness to the people on the days that the ozone level is high, and how people can change their behavior to avoid coming into contact with the gas (for instance, avoiding going outside during the hours that the gas is most likely to be present).

3. Further Research and Analysis:

- Therefore, the current analysis brings out the applicable findings; clearly, more research needs to be done on the impact of other contributing factors like Wind Speed and Temperature on ozone level. Further research could also focus on the day to night variation in the concentration of ozone and its effects on the quality of the atmosphere.
- This dataset could also be analyzed using more complicated methods like regression analysis, or any other machine learning model to predict future ozone levels given environmental factors. These sort of predictive models may be utilized by policymakers to make preventive decisions regarding pollution and other health hazards.

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